

Multi-stress Failure Simulation App

This repository contains the source code for a Streamlit-based visualization app for remaining useful life (RUL) prediction and failure-mechanism simulation under coupled environmental stresses.

中文说明见 [README_zh.md](#).

The implementation uses the NASA C-MAPSS FD004 turbofan degradation dataset as the experimental data source and combines:

- LightGBM-based RUL prediction
- Weibull reliability modeling
- hidden Markov model (HMM) degradation-state simulation
- SHAP-based model interpretation
- Monte Carlo lifetime sampling

Repository Scope

This course-project repository contains the simulation program, the experiment paper, the presentation slides, and the supporting figures used for submission.

Requirements

- Python 3.10+
- Windows, macOS, or Linux
- No GPU required

Install dependencies:

```
pip install -r requirements.txt
```

On high-core-count servers, limiting numerical library threads is recommended:

```
export OPENBLAS_NUM_THREADS=4
export OMP_NUM_THREADS=4
export MKL_NUM_THREADS=4
export LGBM_NUM_THREADS=4
```

Data Preparation

Download or prepare the NASA C-MAPSS FD004 files and place them under data/:

```
data/
├── train_FD004.txt
├── test_FD004.txt
└── RUL_FD004.txt
```

For the current course submission snapshot, the FD004 files are already placed in data/.

Run

Start the visualization app:

```
OPENBLAS_NUM_THREADS=4 OMP_NUM_THREADS=4 MKL_NUM_THREADS=4 LGBM_NUM_THREADS=4 streamlit run
```

For remote access, bind Streamlit to all interfaces:

```
OPENBLAS_NUM_THREADS=4 OMP_NUM_THREADS=4 MKL_NUM_THREADS=4 LGBM_NUM_THREADS=4 \
streamlit run app.py \
  --server.headless true \
  --server.address 0.0.0.0 \
  --server.port 8501 \
  --server.enableCORS false \
  --server.enableXsrfProtection false
```

The first visit loads data and trains the models. Later interactions reuse the cached pipeline.

Build Paper and Slides

Compile the course paper:

```
cd paper
xelatex -interaction=nonstopmode -halt-on-error paper.tex
```

Compile the presentation slides:

```
cd slide
xelatex -interaction=nonstopmode -halt-on-error presentation.tex
```

Features

1. **Weibull Reliability Simulation**
Compare reliability curves under different stress types and stress levels.
2. **Degradation State Evolution**
Visualize HMM state trajectories and forward state-probability prediction for selected units.
3. **RUL Prediction Comparison**
Compare single-stress models with the coupled multi-stress model using line and scatter plots.
4. **Monte Carlo Lifetime Simulation**
Adjust stress multipliers and observe how the sampled lifetime distribution changes.
5. **Model Metrics**
Inspect RMSE, MAE, R^2 , ± 20 -cycle accuracy, and PHM08 score.
6. **SHAP Model Interpretation**
Inspect global feature importance and dependency plots for the coupled model.

Project Structure

```
course_simulation/
├── app.py                # Streamlit app entry point
├── requirements.txt      # Python dependencies
├── README_zh.md         # Chinese project guide
├── SUBMISSION_CHECKLIST.md
├── data/
│   ├── README.md        # Dataset placement instructions
│   ├── train_FD004.txt
│   ├── test_FD004.txt
│   └── RUL_FD004.txt
├── paper/               # Course paper source and PDF
├── slide/               # Presentation source, PDF, and notes
├── scripts/             # Figure-generation helpers
├── src/
│   ├── __init__.py
│   └── engine.py        # Simulation and modeling pipeline
└── results/             # Runtime output directory
```

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